Chapter 24

Capacitance and Dielectrics

Chapter Objectives

- To understand that a capacitor is a device that stores electric potential energy and electric charge.
- To know that capacitance refers to the ratio of the charge on each conductor to the potential difference between the conductors.
- This ratio increases when an insulating material (a dielectric) is present. This is due to a redistribution of charge, called polarization.

Chapter Objectives

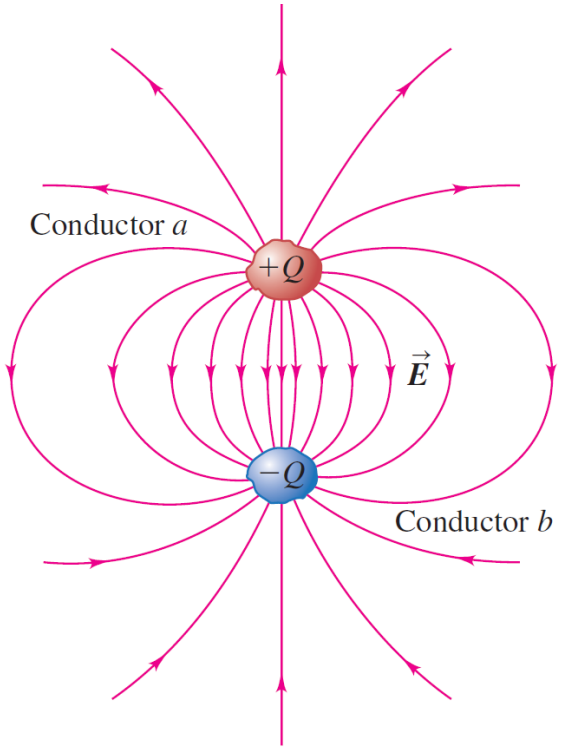
- Studying polarization gives us added insight into the electrical properties of matter, including electric potential energy.

Chapter Outline

1. Capacitors and Dielectrics
2. Capacitors in Series and Parallel
3. Energy Storage in Capacitors and Electric-Field Energy
4. Dielectrics
5. Molecular Model of Induced Charge
6. Gauss's Law In Dielectrics

24.1 Capacitors and Capacitance

- Any two conductors a and b separated by an insulator or a vacuum form a **capacitor**.



24.1 Capacitors and Capacitance

- Electrons are transferred from one conductor to the other, this is called charging the capacitor.
- A capacitor with charge Q (charge Q is stored on the capacitor) means the conductor at higher potential has charge $+Q$ and the conductor at lower potential has charge $-Q$.
- Capacitance is a measure of the ability of a capacitor to store energy.

24.1 Capacitors and Capacitance

- In circuit diagrams a capacitor is represented by either of these symbols: 

$$C = \frac{Q}{V_{ab}} \qquad (24.1)$$

(definition of capacitance)

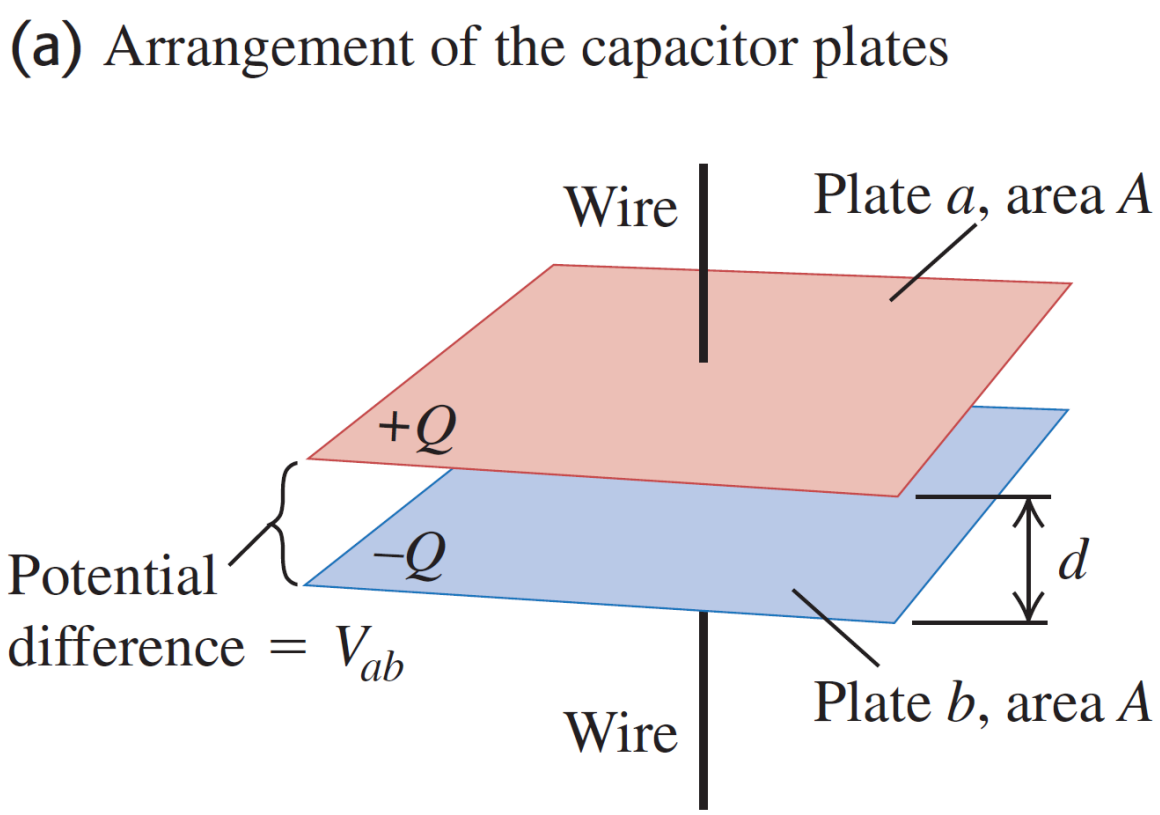
- C: Capacitance
- Q: Magnitude of charge

24.1 Capacitors and Capacitance

- V_{ab}: Fixed potential difference V_{ab} between the conductors (potential of positively charged conductor a with respect to potential of negatively charged conductor b)
- The SI unit of capacitance is called one farad ($1\ \mathrm{F}$).
- $1\ \mathrm{F}$ = 1 farad = $1\ \mathrm{C/V}$ = 1 coulomb/volt

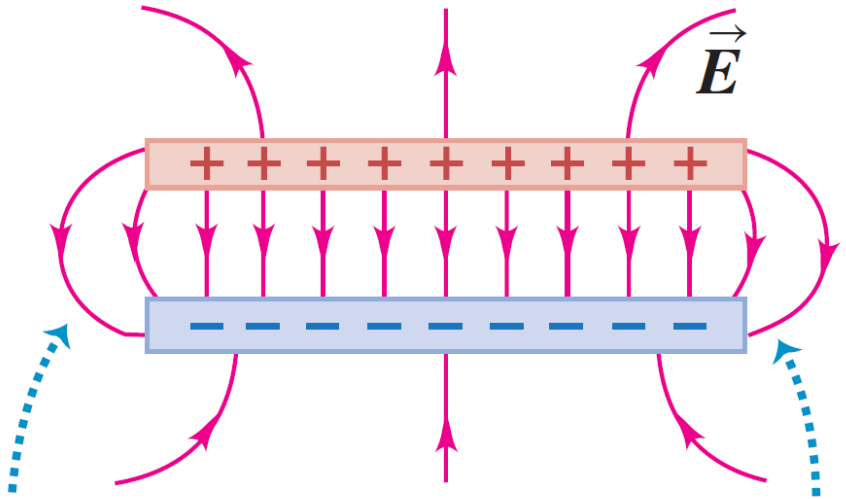
24.1 Capacitors and Capacitance

The simplest form of capacitor consists of two parallel conducting plates, each with area A, separated by a distance d that is small in comparison with their dimensions. We call this arrangement a parallel-plate capacitor.



24.1 Capacitors and Capacitance

(b) Side view of the electric field $\vec{E}$



When the separation of the plates is small compared to their size, the fringing of the field is slight.

24.1 Capacitors and Capacitance

- The field magnitude is

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$$

- The potential difference between the plates is

$$V_{ab} = Ed = \frac{1}{\varepsilon_0}\frac{Qd}{A}$$

24.1 Capacitors and Capacitance

- Therefore the capacitance C of a parallel-plate capacitor in vacuum is

$$C = \frac{Q}{V_{ab}} = \varepsilon_0 \frac{A}{d} \qquad (24.2)$$

(capacitance of a parallel-plate capacitor in vacuum)

- In vacuum, the capacitance C is a constant independent of Q and V_{ab}.

- From Eq. (24.2), $1\ \mathrm{F} = 1\ \mathrm{C}^2/\mathrm{N}\cdot\mathrm{m} = 1\ \mathrm{C}^2/\mathrm{J}$.

24.1 Capacitors and Capacitance

- $\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F/m}$
- One farad is a very large capacitance.
- In many applications the most convenient units of capacitance are the microfarad ($1\ \mu\mathrm{F} = 10^{-6}\ \mathrm{F}$) and the picofarad ($1\ \mathrm{pF} = 10^{-12}\ \mathrm{F}$).

Example 24.1 Size of a $1-\mathrm{F}$ capacitor

A parallel-plate capacitor has a capacitance of $1.0\ \mathrm{F}$. If the plates are $1.0\ \mathrm{mm}$ apart,

what is the area of the plates?

Identify and Set Up

We are given the values of C and d for a parallel-plate capacitor, so we use Eq.(24.2) and solve for the target variable A.

Execute

From Eq. (24.2), the area A is

$$A = \frac{Cd}{\varepsilon_0}$$

$$= \frac{(1.0\ F)(1.0 \times 10^{-3}\ m)}{8.85 \times 10^{-12}\ F/m}$$

$$= 1.1 \times 10^{8}\ m^2$$

Evaluate

This corresponds to a square about $10\,km$ (about $6\,miles$) on a side. It is now possible to make $1-F$ capacitors a few centimeters a side. One type uses activated carbon granules, 1 gram of which has a surface area of about $1000\,m^2$.

Example 24.2 Properties of a parallel-plate capacitor

The plates of a parallel-plate capacitor in vacuum are $5.00\,mm$ apart and $2.00\,m^2$ in area. A potential difference of $10000\,V$ $(10.0\,kV)$ is applied across the capacitor.

Compute

a) the capacitance;

b) the charge on each plate; and

c) the magnitude of the electric field in the space between them.

Identify and Set up

Use the given values of plate area A and plate spacing d in Eq. (24.2) to find the capacitance C.

Then find charge Q on each plate using the given potential difference V_{ab} and Eq. (24.1).

Finally find the electric field between the plates using the relationship $E = \frac{Q}{\varepsilon_0 A}$.

Execute

a) From Eq. (24.2),

$$C = \varepsilon_0 \frac{A}{d} = \frac{(8.85 \times 10^{-12}\ F)(2.00\ m^2)}{5.00 \times 10^{-3}\ m}$$
$$= 3.54 \times 10^{-9}\ F = 0.00354\ \mu F$$

b) The charge on the capacitor is

$$Q = CV_{ab}$$
$$= (3.54 \times 10^{-9}\ C/V)(1.00 \times 10^{4}\ V)$$
$$= 3.54 \times 10^{-5}\ C = 35.4\ \mu C$$

Execute

The plate at higher potential has charge $+35.4\ \mu C$ and the other has charge $-35.4\ \mu C$.

c) The electric-field magnitude is

$$E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$$

$$= \frac{3.54 \times 10^{-5}\ C}{(8.85 \times 10^{-12}\ C^2/N \cdot m^2)(2.00\ m^2)}$$

$$= 2.00 \times 10^6\ N/C$$

Evaluate

An alternative method for c) makes use of the fact that electric field is equal in magnitude to the potential gradient [Eq. (23.22)].

Since the field between the plates is uniform,

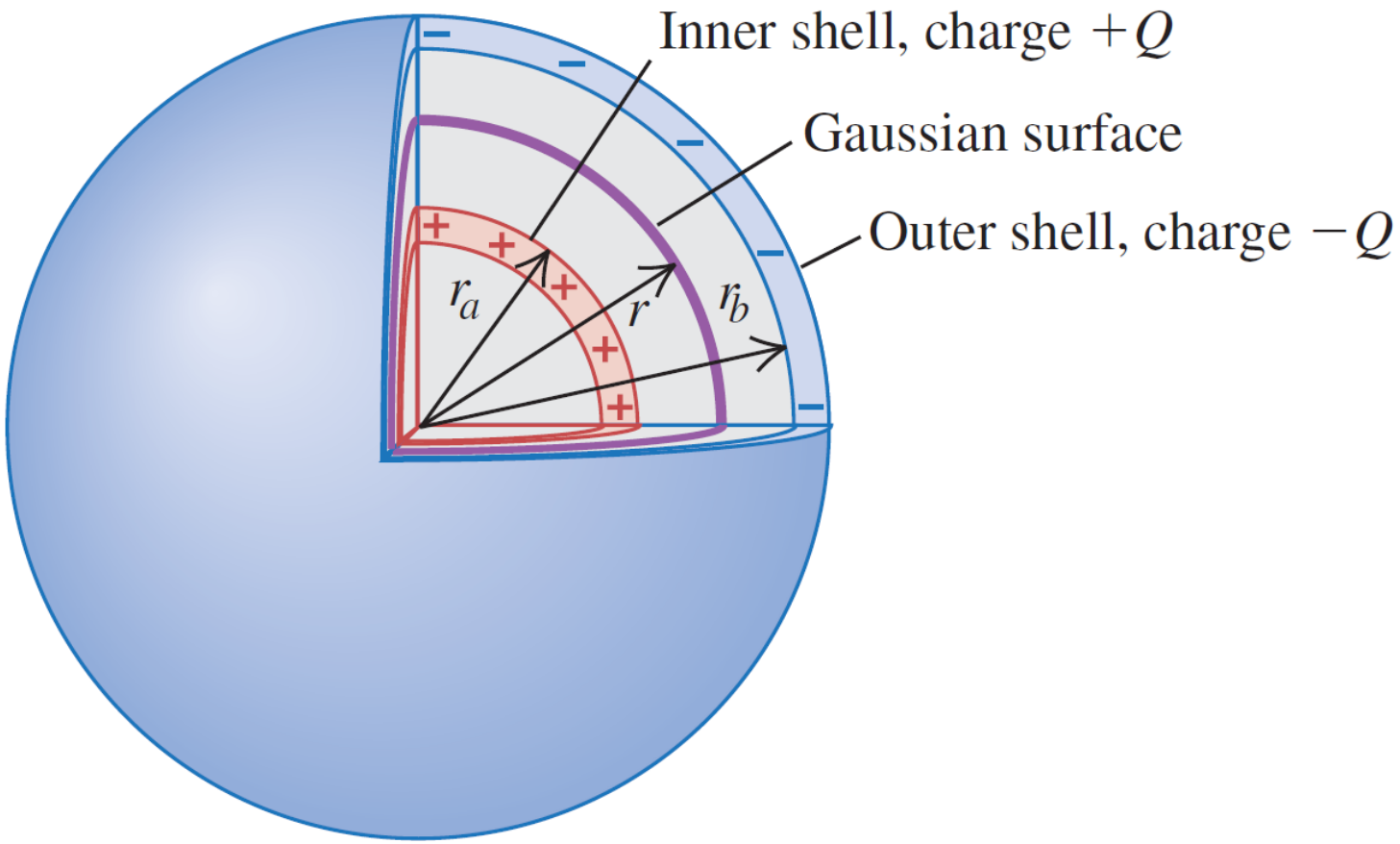
$$E = \frac{V_{ab}}{d} = \frac{1.00 \times 10^4\ V}{5.00 \times 10^{-3}\ m} = 2.00 \times 10^6\ V/m$$

Example 24.3 A spherical capacitor

Two concentric spherical conducting shells are separated by vacuum. The inner shell has total charge $+Q$ and outer radius r_a, and the outer shell has charge $-Q$ and inner radius r_b. (The inner shell is attached to the outer shell by thin insulating rods that have negligible effect on the capacitance.)

Find the capacitance of this spherical capacitor.

Example 24.3 A spherical capacitor



Identify

This is a spherical, not parallel-plate capacitor. Use the fundamental definition of capacitance: the magnitude of the charge on either conductor divided by the potential difference between the conductors.

Set Up

Use Gauss's Law to find the electric field between the spherical conductors.

From this value determine the potential difference V_{ab} between the two conductors; then we use Eq. (24.1) to find the capacitance

$$C = \frac{Q}{V_{ab}}$$

Execute

The electric field between the spheres is just $E = \frac{Q}{4\pi\varepsilon_0 r^2}$ due to the charge on the inner sphere. (The outer sphere has no effect.)

Taking the expression for the potential as the same as for a point charge, i.e. $V = \frac{Q}{4\pi\varepsilon_0 r}$,

$$V_{ab} = V_a - V_b = \frac{Q}{4\pi\varepsilon_0}\frac{r_b - r_a}{r_a r_b}$$

Finally, the capacitance is $C = \frac{Q}{V_{ab}} = 4\pi\varepsilon_0 \frac{r_a r_b}{r_b - r_a}$.

Evaluate

We can relate our expression for C to that for a parallel-plate capacitor. The quantity $4\pi r_a r_b$ is intermediate between the areas $4\pi r_a^2$ and $4\pi r_b^2$ of the two spheres; in fact, it's the geometric mean of the two areas, which we can denote by A_{gm}.

The distance between spheres is $d = r_b - r_a$, so we can write $C = 4\pi\varepsilon_0 \frac{r_a r_b}{r_b - r_a} = \varepsilon_0 \frac{A_{gm}}{d}$. This has the same form as for parallel plates: $C = \varepsilon_0 \frac{A}{d}$.

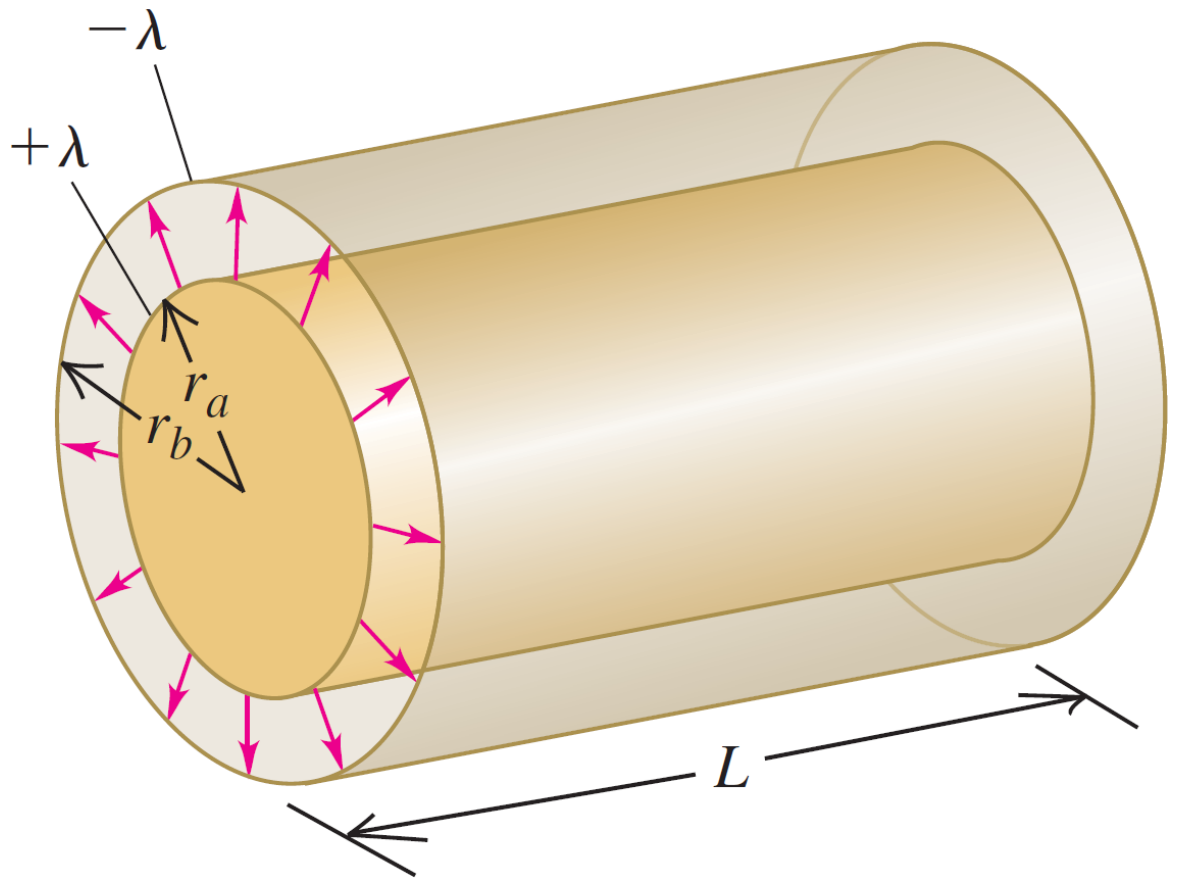
Evaluate

If the distance between spheres is very small in comparison to their radii, they behave like parallel plates with the same area and spacing.

Example 24.4 A cylindrical capacitor

A long cylindrical conductor has a radius r_a and a linear charge density $+\lambda$. It is surrounded by a coaxial cylindrical conducting shell with inner radius r_b and linear charge density $-\lambda$.

Calculate the capacitance per unit length for this capacitor, assuming that there is vacuum in the space between cylinders.



A long cylindrical capacitor. The linear charge density λ is assumed to be positive in this figure. The magnitude of charge in a length L of either cylinder is λL.

Execute

At a point outside a charged cylinder a distance r from the axis, the potential due to the cylinder is

$$V = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_0}{r}$$

Where r_0 is the arbitrary radius at which $V = 0$. The potential V_{ab} of the inner (positive) cylinder a with respect to the outer (negative) cylinder b is thus

$$V_{ab} = \frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_b}{r_a}$$

Execute

This potential difference is positive, since the inner cylinder is at higher potential than the outer.

The total charge $Q = \lambda L$, so from Eq. (24.1) the capacitance C of length L is

$$C = \frac{Q}{V_{ab}} = \frac{\lambda L}{\frac{\lambda}{2\pi\varepsilon_0}\ln\frac{r_b}{r_a}} = \frac{2\pi\varepsilon_0 L}{\ln(r_b/r_a)}$$

Execute

Substituting $\varepsilon_0 = 8.85 \times 10^{-12}\ F/m = 8.85\ pF/m$, we get

$$\frac{C}{L} = \frac{2\pi\varepsilon_0}{\ln(r_b/r_a)} = \frac{55.6\ pF/m}{\ln(r_b/r_a)}$$

Evaluate

The capacitance of the coaxial cylinders is determined entirely by the dimensions, just as for the parallel-plate case.

A typical cable for TV antennas and VCR connections has a capacitance length of $69\ pF/m$.

Chapter 24.2

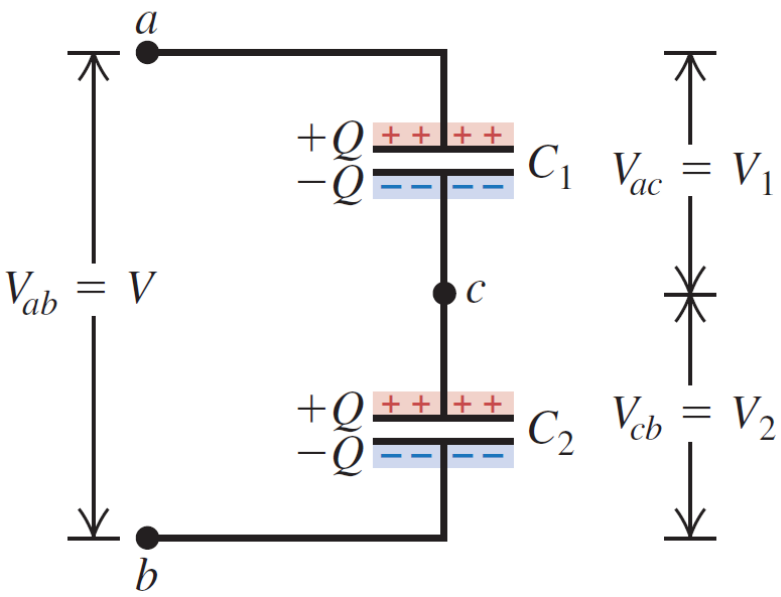
Capacitors in Series and Parallel

24.2 Capacitors in Series and Parallel

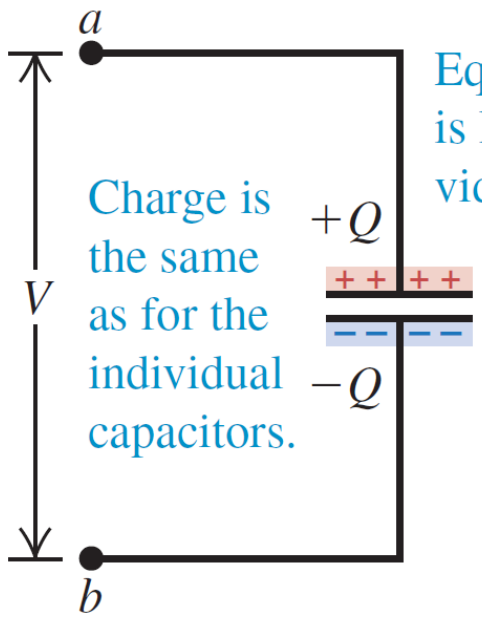
(a) Two capacitors in series

Capacitors in series:

- The capacitors have the same charge Q.
- Their potential differences add: $V_{ac} + V_{cb} = V_{ab}$.



(b) The equivalent single capacitor



Charge is the same as for the individual capacitors.

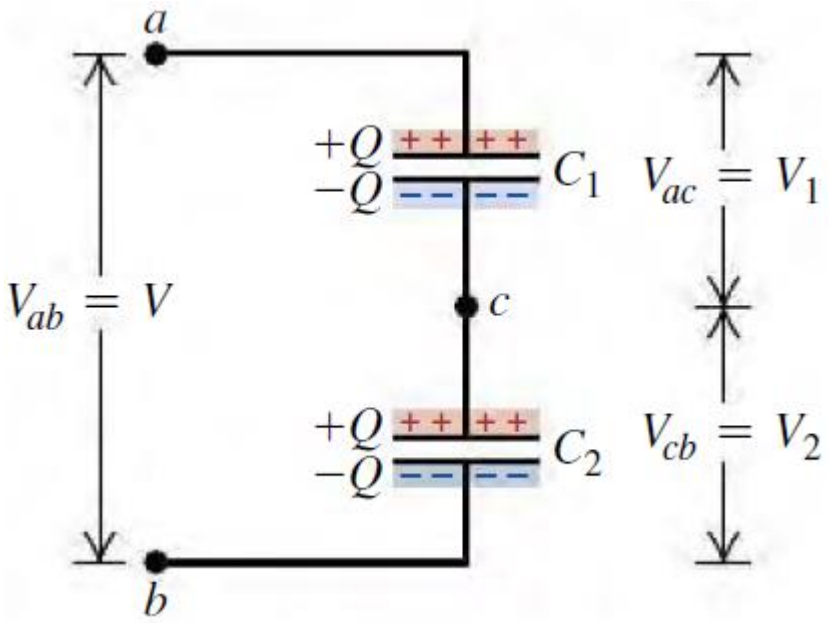
Equivalent capacitance is less than the individual capacitances:

$$C_{eq} = \frac{Q}{V}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

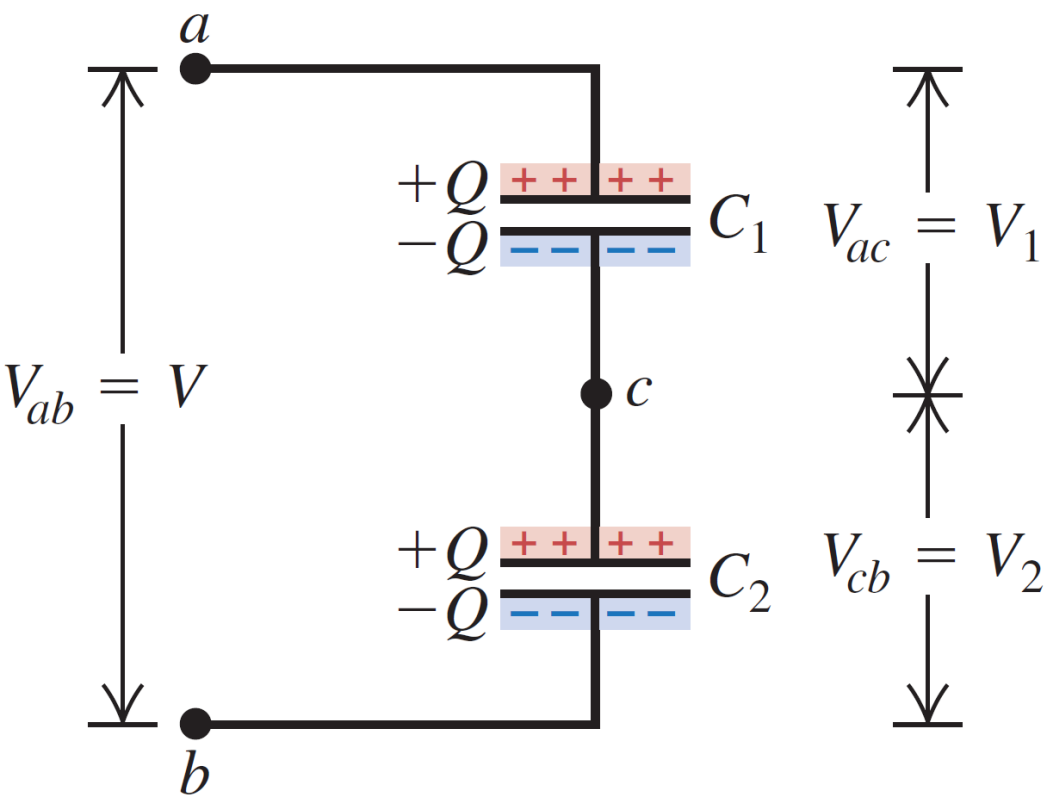
24.2 Capacitors in Series and Parallel

- Fig. a is a schematic diagram of a series connection.
- Both capacitors are initially uncharged.



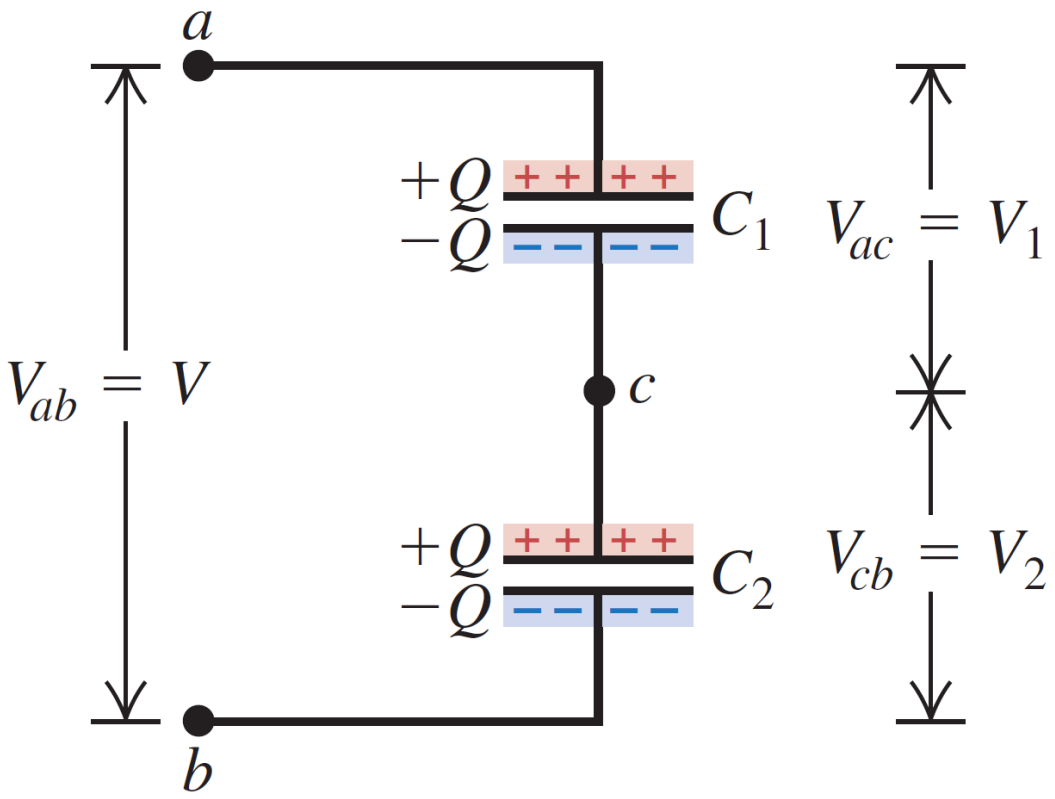
24.2 Capacitors in Series and Parallel

- When a constant positive potential difference V_{ab} is applied between points a and b, the capacitors become charged.
- The figure shows that the charge on all conducting plates have the same magnitude.



24.2 Capacitors in Series and Parallel

- The total charge on the lower plate of C_1 and the upper plate of C_2 together must always be zero because these plates are not connected to anything except each other.



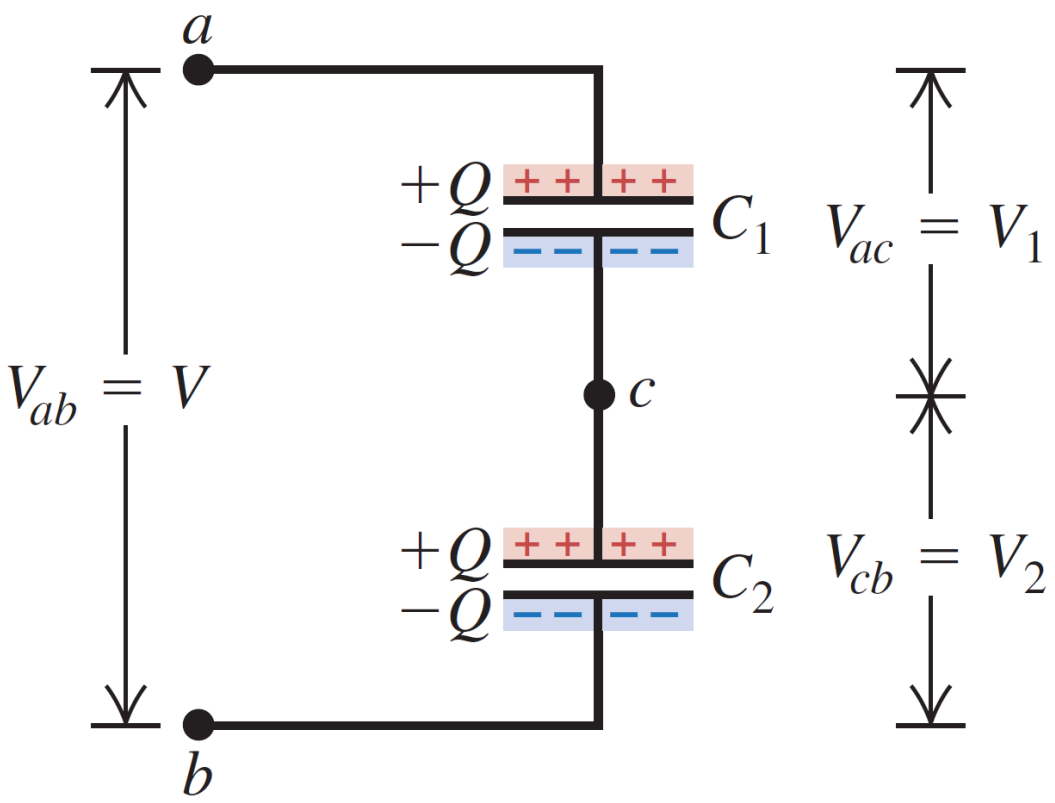
24.2 Capacitors in Series and Parallel

- Thus in a series connection the magnitude of charge on all plates is the same.

Referring to Fig. a:

$$V_{ac} = V_1 = \frac{Q}{C_1} \qquad V_{cb} = V_2 = \frac{Q}{C_2}$$

$$V_{ab} = V = V_1 + V_2 = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right)$$



24.2 Capacitors in Series and Parallel

V_{ac}: denotes the potential difference across the second capacitor

V_{ab}: denotes the potential difference the entire combination of capacitors.

Therefore:

$$\frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} \qquad (24.3)$$

24.2 Capacitors in Series and Parallel

The equivalent capacitance C_{eq} of the series combination is

$$C_{eq} = \frac{Q}{V} \quad or \quad \frac{1}{C_{eq}} = \frac{V}{Q} \qquad (24.4)$$

Combining Eq. (24.3) and (24.4):

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

24.2 Capacitors in Series and Parallel

Extending this analysis to any number of capacitors in series, we obtain the following result:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \quad (capacitors\ in\ series) \qquad (24.5)$$

The reciprocal of the equivalent capacitance of a series combination equals the sum of the reciprocals of the individual capacitances.

In a series connection the equivalent capacitance is always less than any individual capacitance.



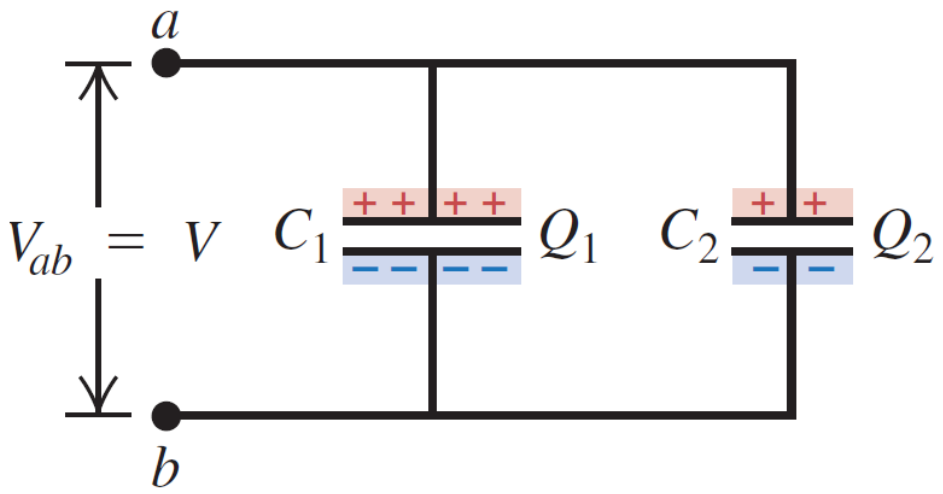
- The magnitude of charge is the same on all plates of the capacitors in a series combination;
- however the potential differences of the individual capacitors are not, unless their individual capacitances are the same.
- The total potential difference across the series combination is the sum of the potential differences of the individual capacitors.

24.2 Capacitors in Series and Parallel

(a) Two capacitors in parallel

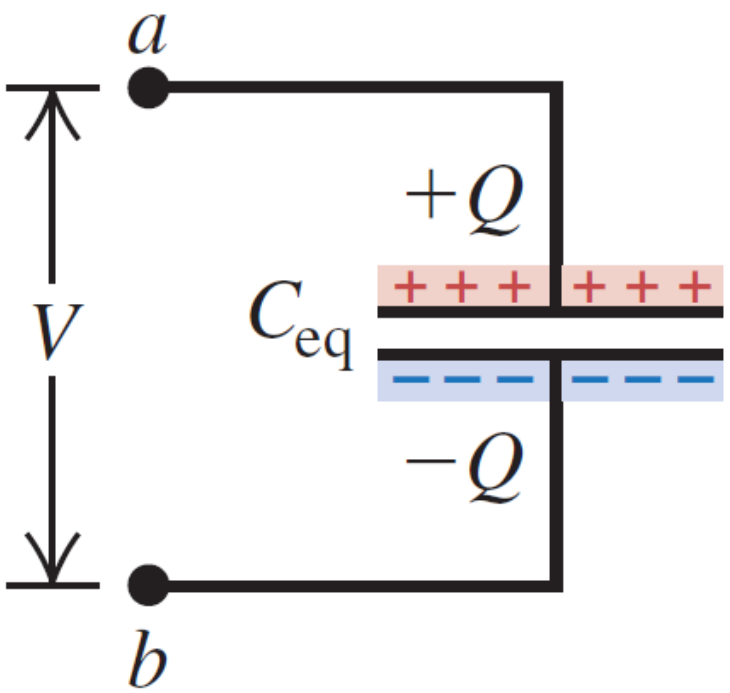
Capacitors in parallel:

- The capacitors have the same potential V.
- The charge on each capacitor depends on its capacitance: $Q_1 = C_1V$, $Q_2 = C_2V$.



24.2 Capacitors in Series and Parallel

(b) The equivalent single capacitor



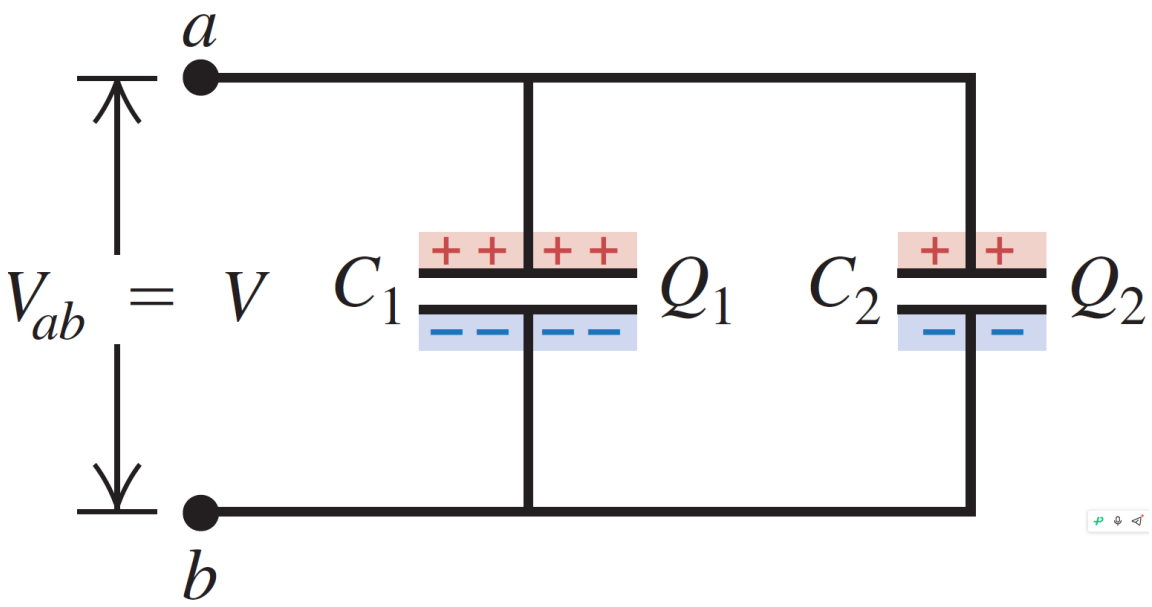
Charge is the sum of the individual charges:

$$Q = Q_1 + Q_2$$

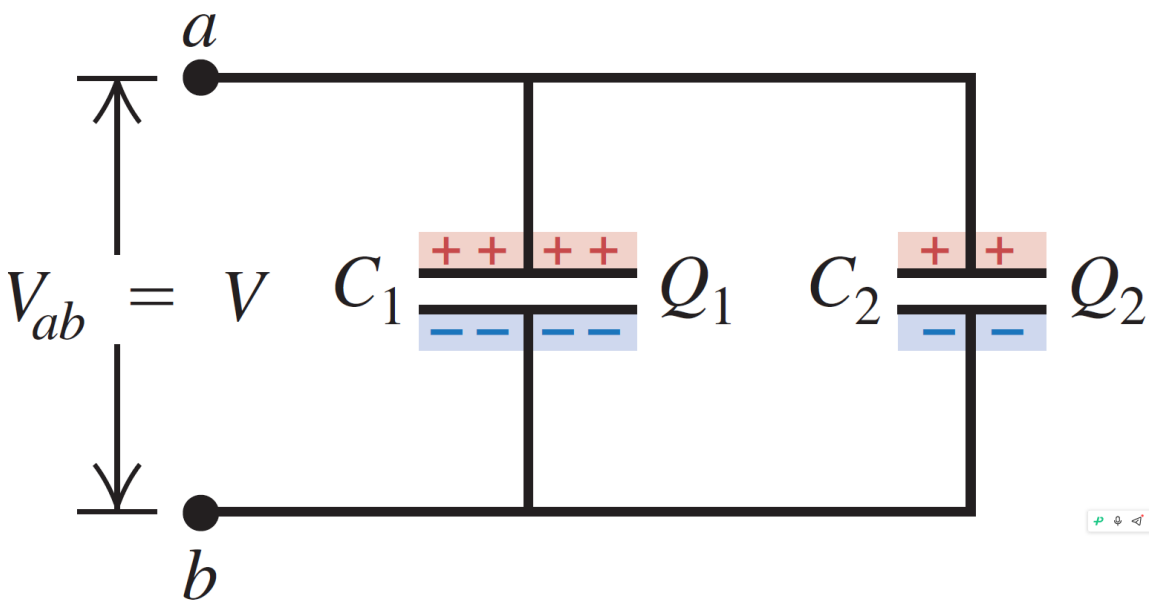
Equivalent capacitance:

$$C_{eq} = C_1 + C_2$$

- The arrangement shown in Fig. a is called a parallel connection.
- The upper plates of the two capacitors are connected together by conducting wires to form an equipotential surface, and the lower plates form another.



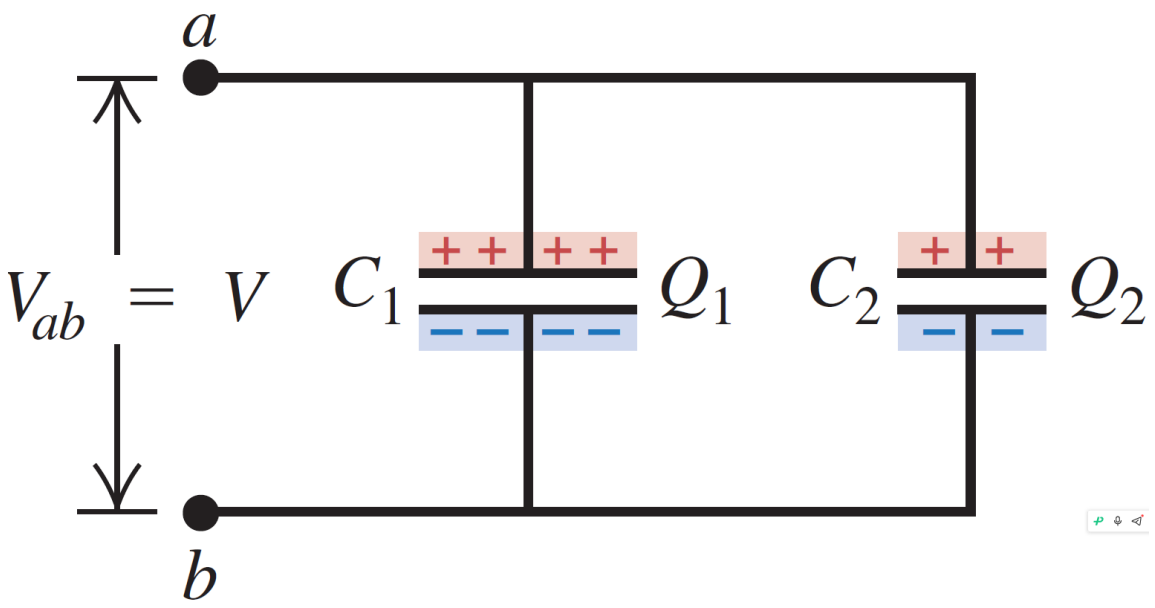
- Hence in a parallel connection the potential difference for all individual capacitors is the same and is equal to $V_{ab} = V$.
- The total charge Q of the combination and thus the total charge on the equivalent capacitor is $Q = Q_1 + Q_2 = (C_1 + C_2)V$



So

$$\frac{Q}{V} = C_1 + C_2 \qquad (24.6)$$

From Eq. (24.6), $C_{eq} = C_1 + C_2$



In the same way we can show that for any number capacitors in parallel,

$$C_{eq} = C_1 + C_2 + C_3 + \cdots \quad (capacitors\ in\ parallel) \qquad (24.7)$$

The equivalent capacitance of a parallel combination equals the sum of the individual capacitances.

In a parallel connection the equivalent capacitance is always greater than any individual capacitance.

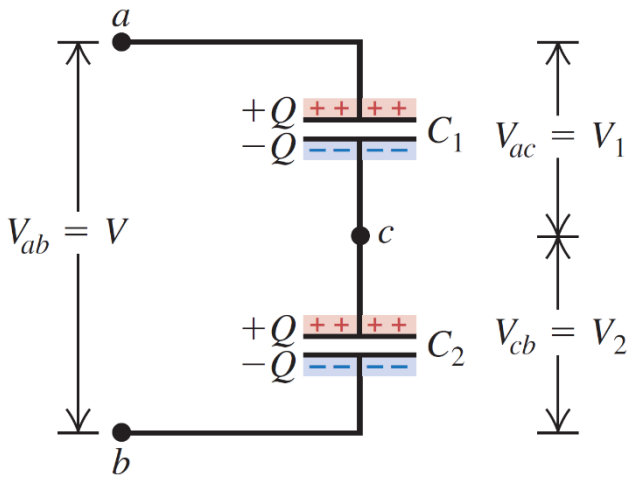


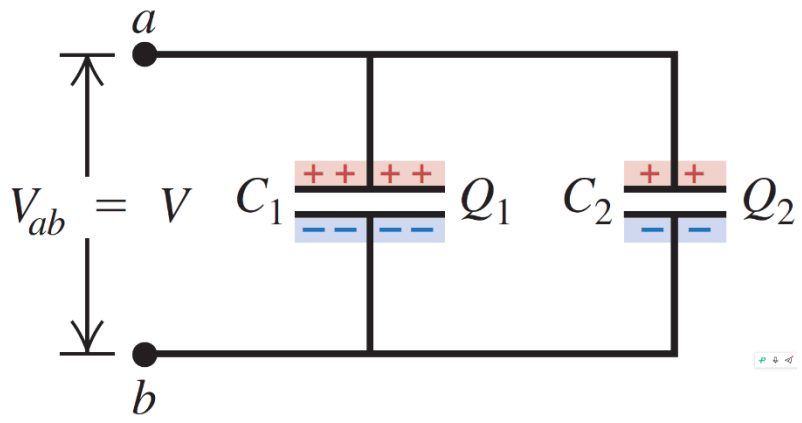
- The potential differences are the same for all the capacitors in a parallel combination;
- however the charges of the individual capacitors are not, unless their individual capacitances are the same.
- The total charge across the series combination is the sum of the charges of the individual capacitors.

Example 24.5 Capacitors in series and in parallel

In the figures below, let $C_1 = 6.0\ \mu F$, $C_2 = 3.0\ \mu F$, and $V_{ab} = 18\ V$.

Find the equivalent capacitance, and the charge and potential difference for each capacitor when the two capacitors are connected a) in series and b) in parallel.





Identify and Set up

The equivalent capacitance C_{eq} for part (a) and part (b) is given by Eq. (24.5) and Eq. (24.6) respectively.

In each part the charge and potential difference from the definition of capacitance.

Execute

(a) For a series combination:

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{6.0\ \mu F} + \frac{1}{3.0\ \mu F}$$

$$C_{eq} = 2.0\ \mu F$$

The charge Q on each capacitor in series is the same as that on the equivalent capacitor:

$$Q = C_{eq}V = (2.0\ \mu F)(18\ V) = 36\ \mu C$$

Execute

The potential difference across each capacitor is inversely proportional to its capacitance:

$$V_{ac} = V_1 = \frac{Q}{C_1} = \frac{36\ \mu C}{6.0\ \mu F} = 6.0\ V$$

$$V_{cb} = V_2 = \frac{Q}{C_2} = \frac{36\ \mu C}{3.0\ \mu F} = 12.0\ V$$

Execute

(b) For a parallel combination:

$$C_{eq} = C_1 + C_2 = 6.0\ \mu F + 3.0\ \mu F = 9.0\ \mu F$$

The potential difference across each of the two capacitors in parallel is the same as across the equivalent capacitor, $18V$. The charge on each capacitor is directly proportional to its capacitance:

$$Q_1 = C_1 V = (6.0\ \mu F)(18\ V) = 108\ \mu C$$

$$Q_2 = C_2 V = (3.0\ \mu F)(18\ V) = 54\ \mu C$$

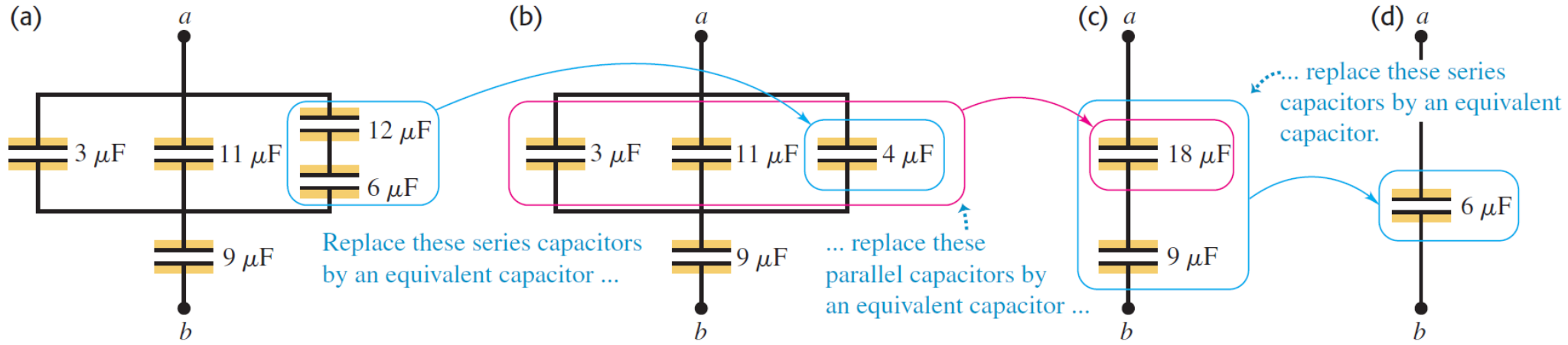
Evaluate

Note that the equivalent capacitance C_{eq} for the series combination in part (a) is indeed less than either C_1 or C_2, while that for the parallel combination in part (b) is greater than either C_1 or C_2.

Example 24.6 A capacitor network

Find the equivalent capacitance of the combination shown in Fig. a below.

24.10 (a) A capacitor network between points a and b. (b) The 12-μF and 6-μF capacitors in series in (a) are replaced by an equivalent 4-μF capacitor. (c) The 3-μF, 11-μF, and 4-μF capacitors in parallel in (b) are replaced by an equivalent 18-μF capacitor. (d) Finally, the 18-μF and 9-μF capacitors in series in (c) are replaced by an equivalent 6-μF capacitor.



Identify and Set up

- Identify portions of the arrangement that are either in series or parallel and combine them to find the net equivalent capacitance.

Execute

- Use Eq. (24.5) to replace the $12\ \mu F$ and $6\ \mu F$ series combination by its equivalent capacitance, C'.

$$\frac{1}{C'} = \frac{1}{12\ \mu F} + \frac{1}{6\ \mu F}$$

$$C' = 4\ \mu F$$

- Use Eq. (24.7) to find the equivalent of the three capacitors in parallel, C'':

$$C'' = 3\ \mu F + 11\ \mu F + 4\ \mu F = 18\ \mu F$$

Execute

- Finally, find the equivalent capacitance of the two capacitors in the equivalent series combination (Fig. 24.8d):

$$\frac{1}{C_{eq}} = \frac{1}{18\ \mu F} + \frac{1}{9\ \mu F}$$

$$C_{eq} = 6\ \mu F$$

Evaluate

- The equivalent capacitance of the network is $6\ \mu F$; if a potential difference of V_{ab} is applied across the terminals of the network, the net charge on the network is $6\ \mu F$ times V_{ab}.

Chapter 24.3

Energy Storage in Capacitors & Electric-Field Energy

24.3 Energy Storage in Capacitors & Electric-Field Energy

- The total work needed to increase the capacitor charge q from zero to a final value Q is

$$W = \int_0^W dW = \frac{1}{C}\int_0^Q q\,dq = \frac{Q^2}{2C} \qquad (24.8)$$

(work to charge a capacitor)

- If we define the potential energy of an uncharged capacitor to be zero, then W in Eq. (24.8) is equal to the potential energy U of the charged capacitor.

24.3 Energy Storage in Capacitors & Electric-Field Energy

- The final stored charge is $Q = CV$, so we can express U (which is equal to W) as

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV \qquad (24.9)$$

(potential energy stored in a capacitor)

- When Q is in coulombs, C in farads (coulombs per volt) and V in volts (joules per coulomb), U is in joules.

24.3 Energy Storage in Capacitors & Electric-Field Energy

- Energy density, denoted by u, is the energy per unit volume in the space between the plates of a parallel-plate capacitor with plate area A and separation d. Using Eq. (24.9),

$$u = Energy\ density = \frac{\frac{1}{2}CV^2}{Ad} \qquad (24.10)$$

- From Eq. (24.2) the capacitance C is given by $C = \varepsilon_0 A/d$. The potential difference V is related to the electric field magnitude E by $V = Ed$.

24.3 Energy Storage in Capacitors & Electric-Field Energy

- Using these expressions in Eq. (24.10), the geometric factors A and d cancel, and we find

$$u = \frac{1}{2}\varepsilon_0 E^2 \qquad (24.11)$$

(electric energy density in a vacuum)

- Eq. (24.11) is valid for any capacitor in vacuum and indeed for any electric field configuration in vacuum.



- Electric-field energy is not different from electric potential energy, just a different way of interpreting it.

- The energy of a given system of charges can be thought of as being a shared property of all the charges or as a property of the electric field created by the charges.

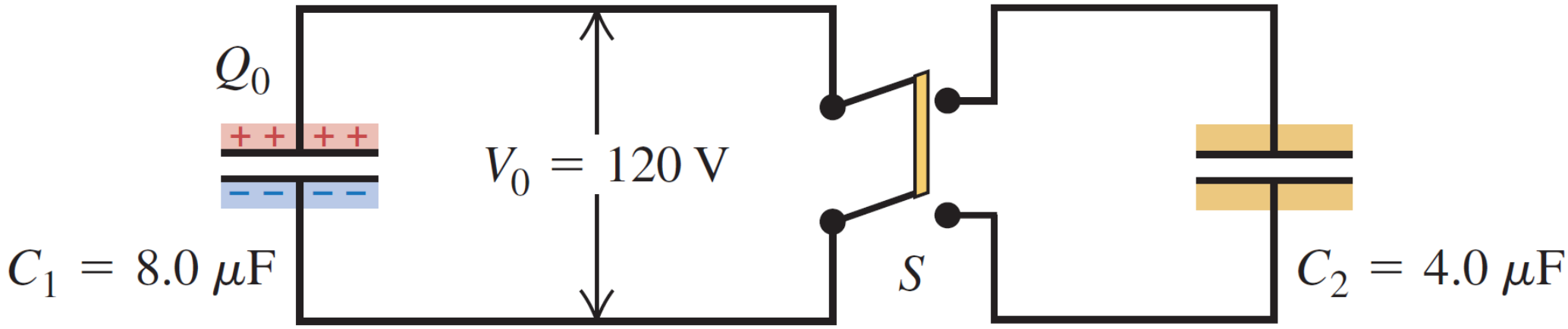
Example 24.7 Transferring charge & energy between capacitors

We connect a capacitor $C_1 = 8.0\,\mu F$ to a power supply, charge it to a potential difference $V_0 = 120\,V$, and disconnect the power supply. Switch S is open.

(a) What is the charge Q_0 on C_1?
(b) What is the energy stored in C_1?
(c) Capacitor $C_2 = 4.0\,\mu F$ is initially uncharged. We close switch S. After charge no longer flows, what is the potential difference across each capacitor, and what is the charge on each capacitor?
(d) What is the final energy of the system?

Example 24.7 Transferring charge & energy between capacitors

24.12 When the switch S is closed, the charged capacitor C_1 is connected to an uncharged capacitor C_2. The center part of the switch is an insulating handle; charge can flow only between the two upper terminals and between the two lower terminals.



Identify and Set Up

- In parts (a) and (b) we find the charge Q_0 and stored energy $U_{initial}$ for the single capacitor C_1 from Eqns. (24.1) and (24.9) respectively.
- After the switch is closed, one wire connects the upper plates of the two capacitors and another wire connects the lower plates, therefore the capacitors are connected in parallel.
- In part (c) we use the character of the parallel connection to determine how Q_0 is shared between the two capacitors.

Identify and Set Up

- In part (d) we use Eq. (24.9) to find the energy stored in capacitors C_1 and C_2; the energy of the system is the sum of these values.
- The total energy is the sum of the energy stored in each capacitor.

Execute

(a) The initial charge Q_0 on C_1 is

$$Q_0 = C_1 V_0 = (8.0\ \mu F)(120\ V) = 960\ \mu C$$

(b) The energy initially stored in C_1 is

$$U_{initial} = \frac{1}{2} Q_0 V_0 = \frac{1}{2}(960 \times 10^{-6}\ C)(120\ V) = 0.058\ J$$

(c) Using conservation of charge,

$$V = \frac{Q_0}{C_1 + C_2} = \frac{960\ \mu C}{8.0\ \mu F + 4.0\ \mu F} = 80\ V$$

Execute

$$Q_1 = 640\,\mu C \qquad Q_2 = 320\,\mu C$$

(d) The final energy of the system is the sum of the energies stored in each capacitor:

$$\begin{aligned} U_{final} &= \frac{1}{2}Q_1V + \frac{1}{2}Q_2V = \frac{1}{2}Q_0V \\ &= \frac{1}{2}(960 \times 10^{-6}\,C)(80\,V) \\ &= 0.038\,J \end{aligned}$$

Evaluate

- The final energy is less than the original energy $U_{initial} = 0.058\,J$.
- The conductors become warmer because of their resistance and some energy is radiated as electromagnetic waves.

Example 24.8 Electric-field energy

Suppose you want to store $1.00\,J$ of electric potential energy in a volume of $1.00\,m^3$ in vacuum.

a) What is the magnitude of the electric field?

b) If the field magnitude is ten times larger, how much energy is stored per cubic meter?

Identify and Set Up

- We use the relationship between the electric-field magnitude E and the energy density u.
- In part (a) we use the given information to find the desired energy density u, then use Eq. (24.11) to find value of E.
- In part (b), Eq. (24.11) tells us how u varies with E.

Execute

a)

$$u = \frac{1.00\ J}{1.00\ m^3} = 1.00 J/m^3$$

$$E = \sqrt{\frac{2u}{\varepsilon_0}}$$

$$= \sqrt{\frac{2(1.00\ J/m^3)}{8.85 \times 10^{-12}\ C^2/N \cdot m^2}}$$

$$= 4.75 \times 10^5\ V/m$$

Execute

b)
Eq. (24.1) shows that u is proportional to E^2. If E increases by a factor of 10, u increases by a factor of 100, and the energy density is $100\ J/m^3$.

Evaluate

- The electric field magnitude found in part (a) is sizeable, corresponding to a potential difference of nearly half a million volts over a distance of one meter.

Example 24.9 Two ways to calculate energy stored in a capacitor

The spherical capacitor described in Example 24.3 (Section 24.1) has charges $+Q$ and $-Q$ on its inner and outer conductors.

Find the electric potential energy stored in the capacitor

a) by using the capacitance C found in Example 24.3 and

b) by integrating the electric-field energy density u.

Identify and Set up

We can determine the energy U stored in a capacitor in two ways: in terms of the work done to put the charges on the two conductors, and in terms of the energy in the electric field between the conductors. The descriptions are equivalent, so they must give us the same result.

we can find the capacitance C and the field magnitude E in the space between the conductors.

Identify and Set up

The electric field is zero inside the inner sphere and is also zero outside the inner surface of the outer sphere, because a Gaussian surface with radius $r < r_a$ or $r > r_b$ encloses zero net charge. Hence the energy density is nonzero only in the space between the spheres, $r_a < r < r_b$.

In part (a) we use $U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$ to find U. In part (b) we use $u = \frac{1}{2}\varepsilon_0 E^2$ to find u, which we integrate over the volume between the spheres to find U.

Execute

a) From Eq. (24.9):

$$U = \frac{Q^2}{2C} = \frac{Q^2}{8\pi\varepsilon_0}\frac{r_b - r_a}{r_a r_b}$$

b) The energy density is nonzero from $r_a < r < r_b$, and is not uniform. Integrating u, the energy per unit volume, over the volume between the inner and outer conducting spheres:

Execute

$$U = \int udV = \int_{r_a}^{r_b} \left(\frac{Q^2}{32\pi^2 \varepsilon_0 r^4} \right) 4\pi r^2 dr$$

$$= \frac{Q^2}{8\pi\varepsilon_0} \int_{r_a}^{r_b} \frac{dr}{r^2} = \frac{Q^2}{8\pi\varepsilon_0} \left(-\frac{1}{r_b} + \frac{1}{r_a} \right)$$

$$= \frac{Q^2}{8\pi\varepsilon_0} \frac{r_b - r_a}{r_a r_b}$$

Evaluate

- The same result for U is obtained with either approach.
- Electric potential energy can be regarded as being associated with either, as in part (a), the charges, or, as in part (b), the field; in both cases, the amount of stored energy is the same.

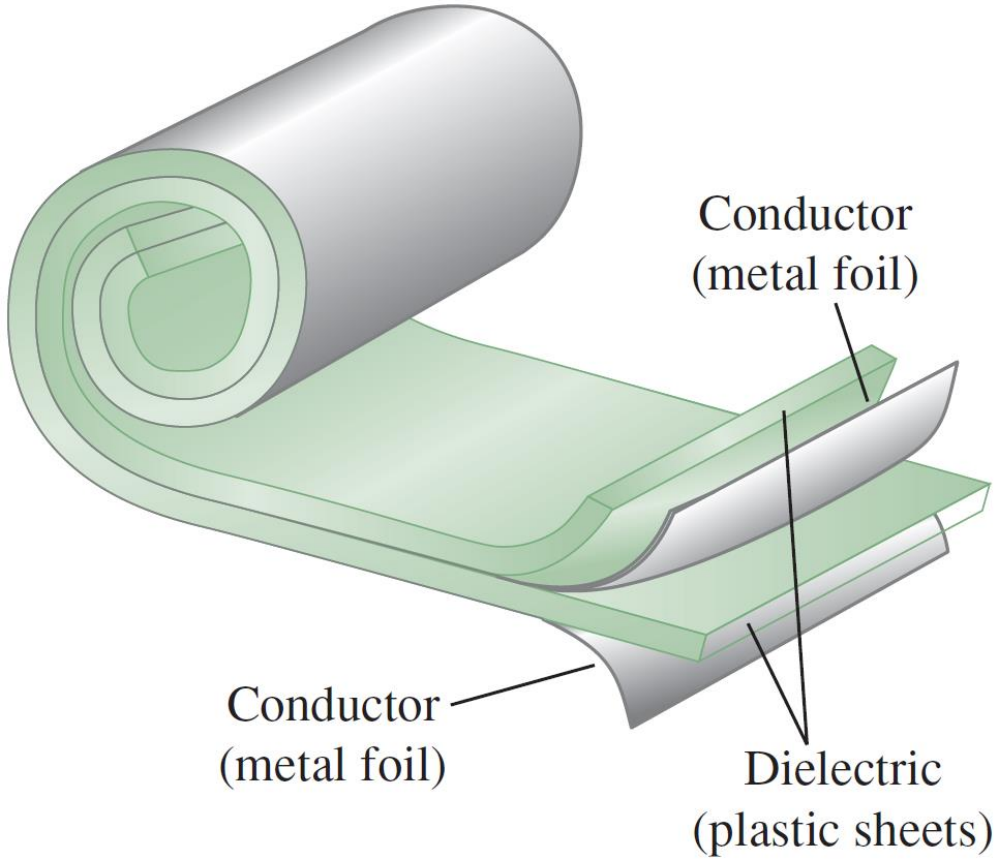
Chapter 24.4

Dielectrics

24.4 Dielectrics

- Most capacitors have a non-conducting material, or dielectric, between their non-conducting plates. (Figure in the next slide)
- A common type of capacitor uses long strips of metal foil for the plates, separated by strips of plastic sheet such as Mylar.
- A sandwich of these materials is rolled up, forming a unit that can provide a capacitance of several microfarads in a compact package.

24.4 Dielectrics



A common type of capacitor uses a dielectric sheet to separate the two conductors.

- Placing a solid dielectric between the plates of a capacitor serves three functions.

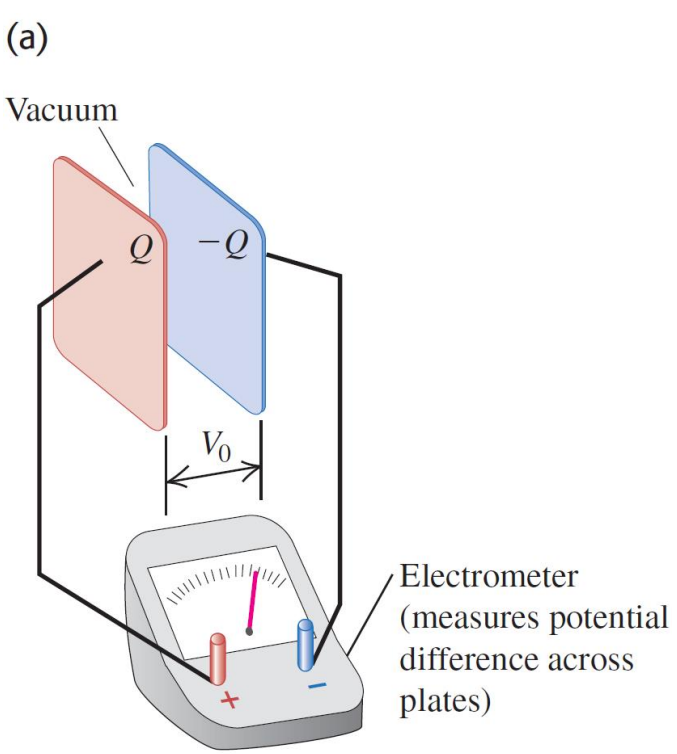
- First, it solves the mechanical problem of maintaining two large metal sheets at a very small separation without actual contact.

- Second, using a dielectric increases the maximum possible potential difference between the capacitor plates. Any insulating material, when subjected to a sufficiently large electric field, experiences a partial ionization that permits conduction through it (Section 23.3). This phenomenon is called **dielectric breakdown**.

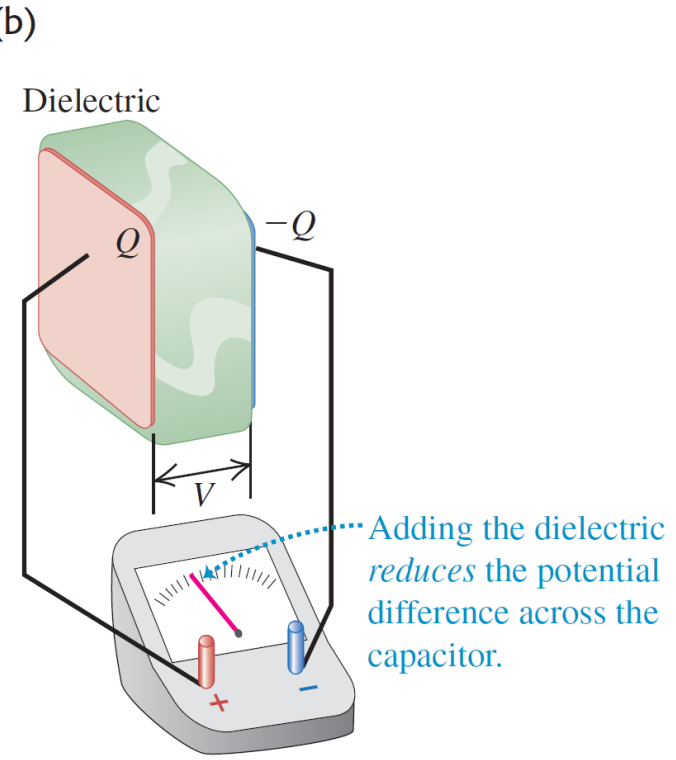
24.4 Dielectrics

- Many dielectric materials can tolerate stronger electric fields without breakdown than can air.
- Thus using a dielectric allows a capacitor to sustain a higher potential difference V and so store greater amounts of charge and energy.
- Third, the capacitance of a capacitor of given dimensions is *greater* when there is a dielectric material between the plates than when there is vacuum.

24.4 Dielectrics



a) With a given charge, the potential difference is V_0



b) With the same charge but with a dielectric between the plates, the potential difference V is smaller than V_0

24.4 Dielectrics

- When the space between plates is completely filled by the dielectric, the ratio of C to C_0 (equal to the ratio of V_0 to V) is called the dielectric constant of the material, K:

$$K = \frac{C}{C_0} \qquad (24.12)$$

(definition of dielectric constant)

24.4 Dielectrics

- When the charge is constant, Eq. (24.12) can be rewritten as

$$V = \frac{V_0}{K} \qquad (24.13)$$

(When Q is constant)

- With the dielectric present, the potential difference for a given charge Q is reduced by a factor K.

24.4 Dielectrics

Table 24.1 Values of Dielectric Constant K at 20°C

Material	K	Material	K
Vacuum	1	Polyvinyl chloride	3.18
Air (1 atm)	1.00059	Plexiglas®	3.40
Air (100 atm)	1.0548	Glass	5–10
Teflon	2.1	Neoprene	6.70
Polyethylene	2.25	Germanium	16
Benzene	2.28	Glycerin	42.5
Mica	3–6	Water	80.4
Mylar	3.1	Strontium titanate	310

24.4 Dielectrics

- When a dielectric material is inserted between the plates while the charge is kept constant, the potential difference between the plates decreases by a factor K. Therefore the electric field between the plates must decreases by the same factor. If E_0 is the vacuum value and E is the value with the dielectric, then

$$E = \frac{E_0}{K} \qquad (24.14)$$

(when Q is constant)

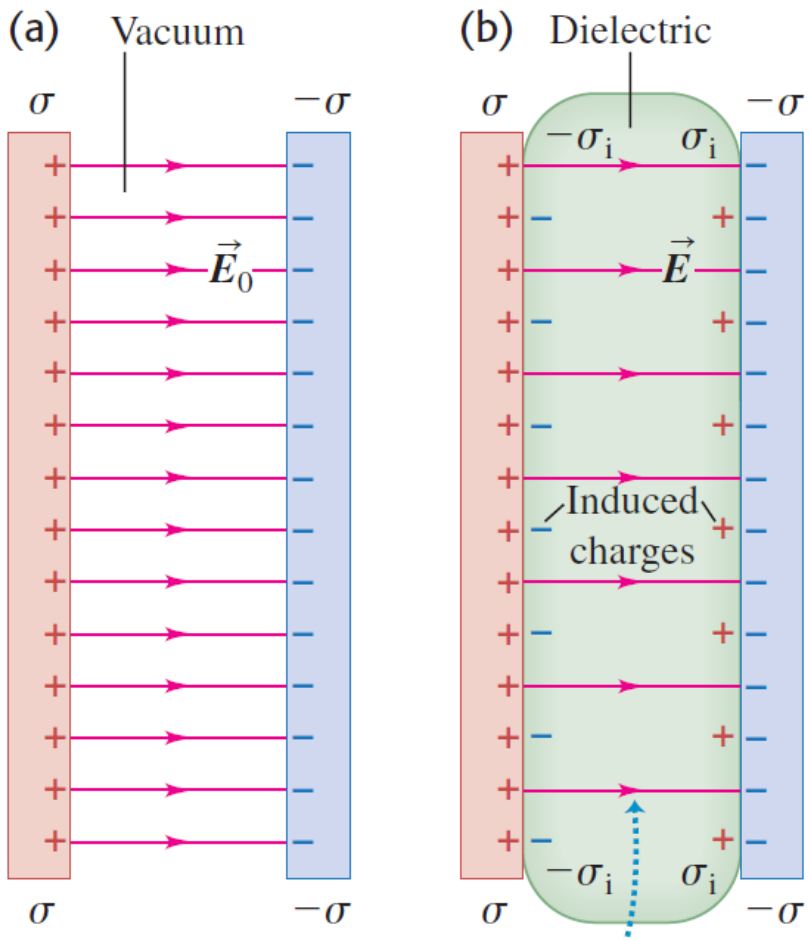
24.4 Dielectrics

- The surface charge on the conducting plates does not change, but an *induced* charge of the opposite sign appears on each surface of the dielectric.
- The dielectric was originally electrically neutral and is still neutral.
- The induced surface charges arise as a result of redistribution of positive and negative charge within the dielectric material, a phenomenon called polarization.

24.4 Dielectrics

(a) Electric filed lines with vacuum between the plates

(b) The induced charges on the faces of the dielectric decrease the electric field.



For a given charge density σ, the induced charges on the dielectric's surfaces reduce the electric field between the plates.

24.4 Dielectrics

- With and without the dielectric, respectively, we have

$$E_0 = \frac{\sigma}{\varepsilon_0} \qquad E = \frac{\sigma - \sigma_i}{\varepsilon_0} \qquad (24.15)$$

- Thus, we find

$$\sigma_i = \sigma\left(1 - \frac{1}{K}\right) \qquad (24.16)$$

(induced surface charge density)

24.4 Dielectrics

- The permittivity of dielectric is

$$\varepsilon = K\varepsilon_0 \qquad (24.17)$$

(definition of permittivity)

- This leads to:

$$E = \frac{\sigma}{\varepsilon} \qquad (24.18)$$

24.4 Dielectrics

- The capacitance when the dielectric is present is given by:

$$C = KC_0 = K\varepsilon_0 \frac{A}{d} = \varepsilon \frac{A}{d} \qquad (24.19)$$

- Repeating the derivation of Eq. (24.11) for the energy density u in an electric field (dielectric present):

$$u = \frac{1}{2} K\varepsilon_0 E^2 = \frac{1}{2} \varepsilon E^2 \qquad (24.20)$$

(electric energy density in a dielectric)

Example 24.10 A capacitor with and without a dielectric

Suppose the parallel plates in Fig. 24.13 each have an area of $2000\ cm^2$ $(2.00 \times 10^{-1}\ m^2)$ and are $1.00\ cm$ $(1.00 \times 10^{-2}\ m)$ apart. The capacitor is connected to a power supply and charged to a potential difference $V_0 = 3000\ V = 3.00\ kV$. It is then disconnected from the power supply, and a sheet of insulating plastic between them. We find that the potential difference decreases to $1000V$ while the charge on each capacitor plate remains constant.

Example 24.10 A capacitor with and without a dielectric

Compute

a) the original capacitance C_0
b) the magnitude of charge Q on each plate
c) the capacitance C after the dielectric is inserted
d) the dielectric constant K of the dielectric
e) the permittivity ε of the dielectric
f) the magnitude of the induced charge Q_i on each face of the dielectric
g. the original electric field E_0 between the plates and
h. the electric field E after the dielectric is inserted.

Example 24.10 A capacitor with and without a dielectric

Identify and Set up

Most of the target variables can be obtained in several different ways. The methods used below are a representative sample.

Execute

a) With vacuum between the plates, we use Eq. (24.19) with $K = 1$:

$$C_0 = \varepsilon_0 \frac{A}{d}$$

$$= (8.85 \times 10^{-12}\ F/m)\frac{2.00 \times 10^{-1}\ m^2}{1.00 \times 10^{-2}\ m}$$

$$= 1.77 \times 10^{-10}\ F$$

$$= 177\ pF$$

Execute

b) From the definition of capacitance,

$$\begin{aligned} Q &= C_0 V_0 \\ &= (1.77 \times 10^{-10}\ F)(3.00 \times 10^{3}\ V) \\ &= 5.31 \times 10^{-7}\ C = 0.531\ \mu C \end{aligned}$$

Execute

c) When the dielectric is inserted, Q is unchanged but the potential difference decreases to $V = 1.00\,kV$. Hence from Eq. (24.1), the new capacitance is

$$C = \frac{Q}{V}$$

$$= \frac{5.31 \times 10^{-7}\ C}{1.0 \times 10^{3}\ V}$$

$$= 531\,pF$$

Execute

d) From Eq. (24.12), the dielectric constant is

$$K = \frac{C}{C_0} = \frac{5.31 \times 10^{-10}\ F}{1.77 \times 10^{-10}\ F} = \frac{531\ pF}{177\ pF} = 3.00$$

Alternatively, from Eq. (24.13),

$$K = \frac{V_0}{V} = \frac{3000\ V}{1000\ V} = 3.00$$

Execute

e) With K from part (d) in Eq. (24.17), the permittivity is

$$\begin{aligned}\varepsilon &= K\varepsilon_0 \\ &= (3.00)(8.85 \times 10^{-12}\ C^2/N \cdot m^2) \\ &= 2.66 \times 10^{-11}\ \ C^2/N \cdot m^2\end{aligned}$$

Execute

f) Multiplying both sides of Eq. (24.15) by the plate area A of each plate gives the induced charge $Q_i = \sigma_i A$ in terms of the charge $Q = \sigma A$ on each plate:

$$Q_i = Q\left(1 - \frac{1}{K}\right)$$

$$= (5.31 \times 10^{-7}\ C)\left(1 - \frac{1}{3.00}\right)$$

$$= 3.54 \times 10^{-7}\ C$$

Execute

g) Since the electric field between the plates is uniform, its magnitude is the potential difference divided by the plate separation:

$$E_0 = \frac{V_0}{d} = \frac{3000\ V}{1.00 \times 10^{-2}\ m} = 3.00 \times 10^5\ V/m$$

h) After the dielectric is inserted,

$$E = \frac{V}{d} = \frac{1000\ V}{1.00 \times 10^{-2}\ m} = 1.00 \times 10^5\ V/m$$

Execute

Or,

$$E = \frac{\sigma}{\varepsilon}$$

$$= \frac{Q}{\varepsilon A}$$

$$= \frac{5.31 \times 10^{-7}\ C}{(2.66 \times 10^{-11} C^2/N \cdot m^2)(2.00 \times 10^{-1}\ m^2)}$$

$$= 1.00 \times 10^5\ V/m$$

Execute

Or,

$$E = \frac{\sigma - \sigma_i}{\varepsilon_0}$$

$$= \frac{Q - Q_i}{\varepsilon_0 A}$$

$$= \frac{(5.31 - 3.54) \times 10^{-7}\ C}{(8.85 \times 10^{-12}\ C^2/N \cdot m^2)(2.00 \times 10^{-1}\ m^2)}$$

$$= 1.00 \times 10^5\ V/m$$

Execute

Or,

$$E = \frac{E_0}{K} = \frac{3.00 \times 10^5\ V/m}{3.00} = 1.00 \times 10^5\ V/m$$

Evaluate

- Our results show that inserting the dielectric increased the capacitance by a factor of $K = 3.00$ and reduced the electric field between the plates by a factor of $1/K = 1/3.00$.
- It did so by developing induced charges on the faces of the dielectric of magnitude $Q(1 - 1/K) = Q(1 - 1/3.00) = 0.667\,Q$.

Example 24.11 Energy storage with and without a dielectric

Find the total energy stored in the electric field of the capacitor in Example 24.10 and the energy density, both before and after the dielectric sheet is inserted.

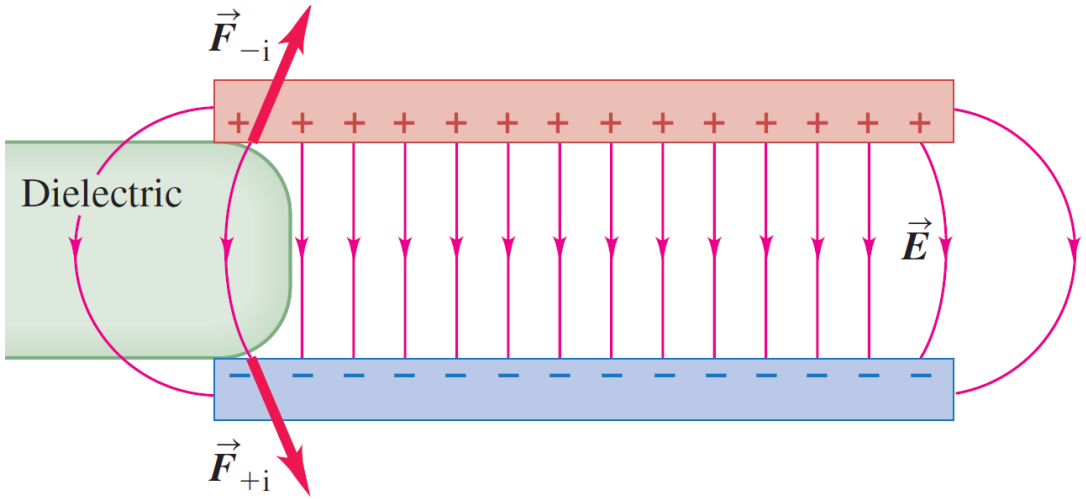


Figure: the fringing field at the edges of the capacitor exerts forces $\vec{F}_{-i}$ and $\vec{F}_{i}$ on the negative and positive induced surface charges of a dielectric, pulling the dielectric into the capacitor.

Identify and Set up

- We use Eq. (24.9) to find the stored energy before and after the dielectric is inserted, and Eq.(24.20) to find the energy density.

Execute

Let the original energy be U_0 and let the energy with the dielectric in place be U. From Eq.(24.9),

$$U_0 = \frac{1}{2} C_0 {V_0}^2$$

$$= \frac{1}{2}(1.77 \times 10^{-10}\ F)(3000\ V)^2 = 7.97 \times 10^{-4}\ J$$

$$U = \frac{1}{2} C V^2$$

$$= \frac{1}{2}(5.31 \times 10^{-10}\ F)(1000\ V)^2 = 2.66 \times 10^{-4}\ J$$

Execute

- The final energy is one-third of the original energy.
- The energy density without the dielectric is given by Eq. (24.20) with $K = 1$:

$$u_0 = \frac{1}{2}\varepsilon_0 {E_0}^2$$
$$= \frac{1}{2}(8.85 \times 10^{-12}\ C^2/N \cdot m^2)(3.00 \times 10^5\ V/m)^2$$
$$= 0.398\ J/m^3$$

Execute

With the dielectric in place,

$$u = \frac{1}{2}\varepsilon E^2$$
$$= \frac{1}{2}(2.66 \times 10^{-11}\ C^2/N \cdot m^2)(1.00 \times 10^5\ V/m)^2$$
$$= 0.133\ J/m^3$$

The energy density with the dielectric is one-third of the original energy density.

Evaluate

- As shown in Fig. 24.14, the fringing field at the edges of a real parallel-plate capacitor tends to pull the dielectric into the space between the plates, doing work on it as it does so.

- We could attach a spring to the bottom of the dielectric in Fig. 24.14 and use this force to stretch the spring.

- Because work is done by the field, the field energy density decreases.

24.4 Dielectrics

- When any dielectric material is subjected to a sufficiently strong electric field, dielectric breakdown takes place and the dielectric becomes a conductor (Fig. 24.15).

- Thus, capacitors always have maximum voltage ratings.

- The maximum electric-field magnitude that a material can withstand without the occurrence of breakdown is called its dielectric strength.

24.4 Dielectrics

- This quantity is affected significantly by temperature, trace impurities and other hard-to-control factors.

- The following table shows the dielectric constant and dielectric strength of some insulating materials.

24.4 Dielectrics

Note that the dielectric strength of dry air is about $3 \times 10^6\ V/m$.

TABLE 24.2 Dielectric Constant and Dielectric Strength of Some Insulating Materials

Material	Dielectric Constant, K	Dielectric Strength, E_m (V/m)
Polycarbonate	2.8	3×10^7
Polyester	3.3	6×10^7
Polypropylene	2.2	7×10^7
Polystyrene	2.6	2×10^7
Pyrex glass	4.7	1×10^7

Concept Summary

Capacitors and capacitance

- A capacitor is any pair of conductors separated by an insulating material.

- When the capacitor is charged, there are charges of equal magnitude Q and opposite sign on the two conductors, and the potential V_{ab} of the positively charged conductor with respect to the negatively charged conductor is proportional to Q.

Capacitors and capacitance

- The capacitance C is defined as the ration of Q to V_{ab}. The SI unit of capacitance is the farad (F): $1\,F = 1\,C/V$.
- A parallel-plate capacitor consists of two parallel conducting plates, each with area A, separated by a distance d.
- If then are separated by vacuum, the capacitance depends on only A and d.

Capacitors and capacitance

- For other geometries, the capacitance can be found by using the definition $C = Q/V_{ab}$.

$$C = \frac{Q}{V_{ab}} \qquad (24.1)$$

(definition of capacitance)

$$C = \frac{Q}{V_{ab}} = \varepsilon_0 \frac{A}{d} \qquad (24.2)$$

(capacitance of a parallel-plate capacitor in vacuum)

Capacitors in series and parallel

- When capacitors with capacitances C_1, C_2, C_3,... are connected in series, the reciprocal of the equivalent capacitance C_{eq} equals the sum of the reciprocals of the individual capacitances.

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \quad (24.5)$$

$$(capacitors\ in\ series)$$

Capacitors in series and parallel

- When capacitors are connected in parallel, the equivalent capacitance C_{eq} equals the sum of the individual capacitances.

$$C_{eq} = C_1 + C_2 + C_3 + \cdots \quad (24.7)$$

$$(capacitors\ in\ parallel)$$

Energy in a capacitor

- The energy U required to charge a capacitor C to a potential difference V and a charge Q is equal to the energy stored in the capacitor.

- This energy can be thought of as residing in the electric field between the conductors

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV \qquad (24.9)$$

(potential energy stored in a capacitor)

Energy in a capacitor

- The energy density u (energy per unit volume) is proportional to the square of the electric-field magnitude.

$$u = \frac{1}{2}\varepsilon_0 E^2 \qquad (24.11)$$

(electric energy density in a vacuum)

Dielectrics

- When the space between the conductors is filled with a dielectric material, the capacitance increases by a factor K, the dielectric constant of the material.

- The quantity $\varepsilon = K\varepsilon_0$ is the permittivity of the dielectric.

$$\varepsilon = K\varepsilon_0 \quad (24.17)$$

(definition of permittivity)

Dielectrics

- For a fixed amount of charge on the capacitor plates, induced charges on the surface of the dielectric decrease the electric field and potential difference between the places by the same factor K.

- The surface charge results from polarization, a microscopic rearrangement of charge in the dielectric.

Dielectrics

- Under sufficiently strong electric fields, dielectrics become conductors, a situation called dielectric breakdown.
- The maximum field that a material can withstand without breakdown is called its dielectric strength.
- In a dielectric, the expression for the energy density is the same as in vacuum but with ε_0 replaced by $\varepsilon = K\varepsilon_0$.

Dielectrics

$$C = KC_0 = K\varepsilon_0 \frac{A}{d} = \varepsilon \frac{A}{d} \qquad (24.19)$$

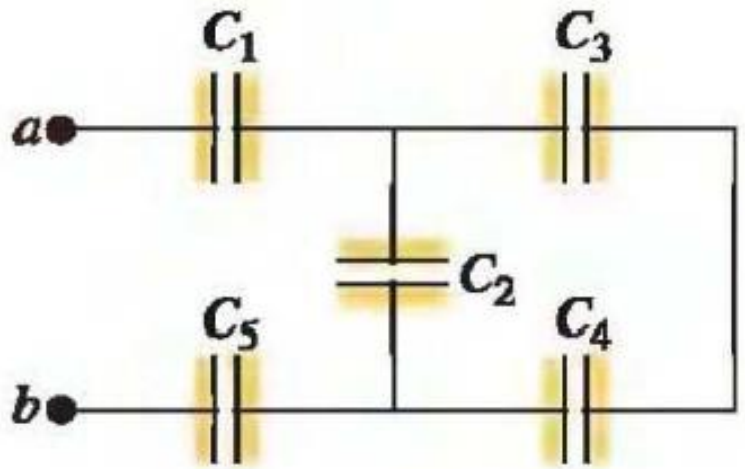
(parallel-plate capacitor, dielectric between plates)

$$u = \frac{1}{2} K\varepsilon_0 E^2 = \frac{1}{2} \varepsilon E^2 \qquad (24.20)$$

(electric energy density in a dielectric)

24.59. In Fig. 24-32, $C_1 = C_5 = 8.4\mu F$ and $C_2 = C_3 = C_4 = 4.2\mu F$. The applied potential is $V_{ab} = 220V$. (a) What is the equivalent capacitance of the network between points a and b? (b) Calculate the charge on each capacitor and the potential difference cross each capacitor.

Figure 24.32 Problem 24.59.



24.71, 2. A parallel-plate capacitor has the space between the plates filled with two slabs of dielectric, one with constant K_1 and one with constant K_2 (Fig. 24.37). Each slab has thickness d/2, where d is the plate separation. Show that the capacitance is

$$C = \frac{2\epsilon_0 A}{d}\left(\frac{K_1 K_2}{K_1 + K_2}\right)$$

Figure 24.37
Problem 24.71.

